

Methods of Power Transmission

- Mechanical power can be transmitted from one shaft to another by following methods
 - Belt drive
 - Rope drive
 - Chain drive
 - Gear drive

The shaft from which power is transmitted is called **driver** shaft and the shaft to which power is transmitted is called **driven shaft**

The choice of the type of drive for power transmission

depend on many factors such as

- Distance between shafts
- Amount of power to be transmitted
- Speed ratio of shafts
- Accuracy required

BELT DRIVE

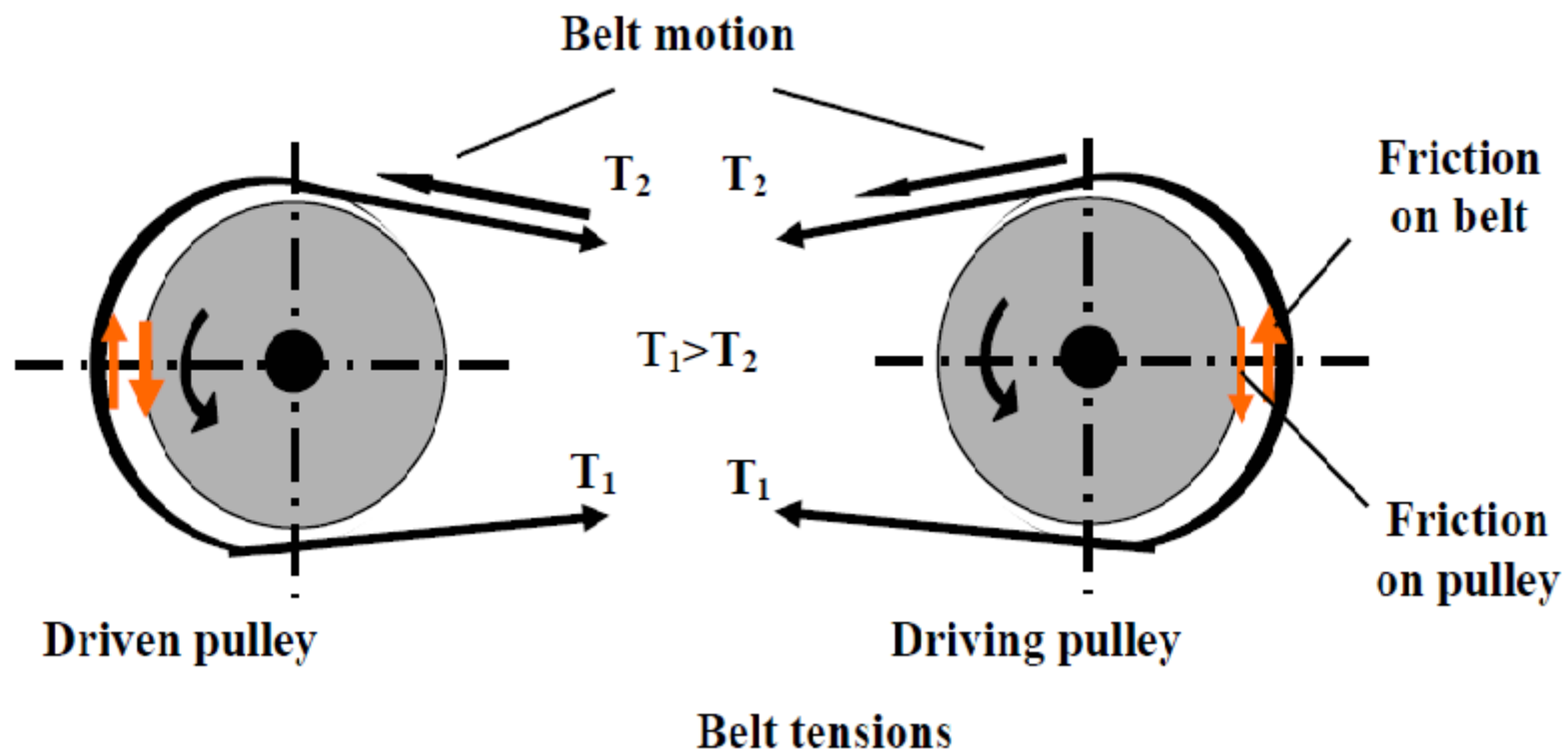
- A belt is a thin inextensible band made of leather, rubber, steel
- Belts are used to transmit power between two parallel shafts, which are at a considerable distance apart
- Belts are made endless to run over the pulleys mounted on the shafts
- Friction between the belt and the pulley is responsible for transmitting power from one pulley to other
- It's a friction drive

BELT DRIVE

- It is not a positive drive, since there is always some possibility of slipping between the belt and pulley

Amount of power transmitted depends on

- Velocity of the belt
- The tension with which the belt is placed under the pulleys
- The arc of contact between the belt and the smaller pulley



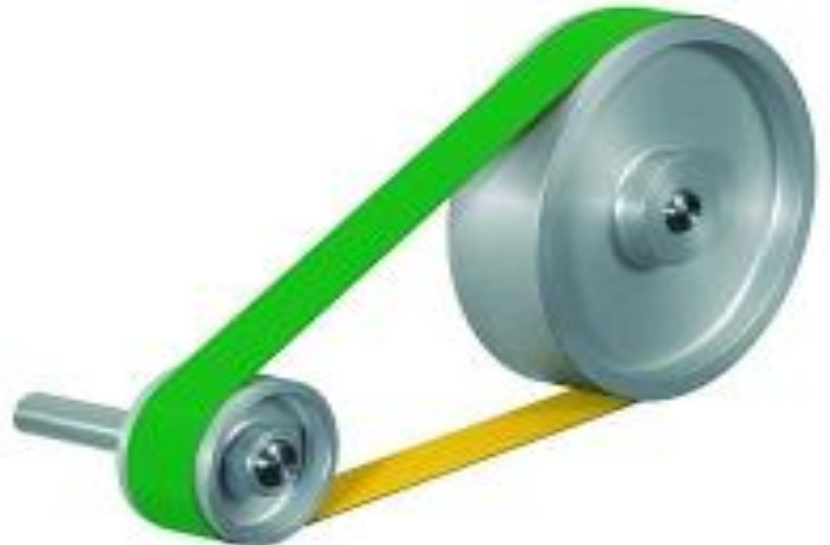
FLAT BELT

V BELT

**TYPES OF
BELTS**

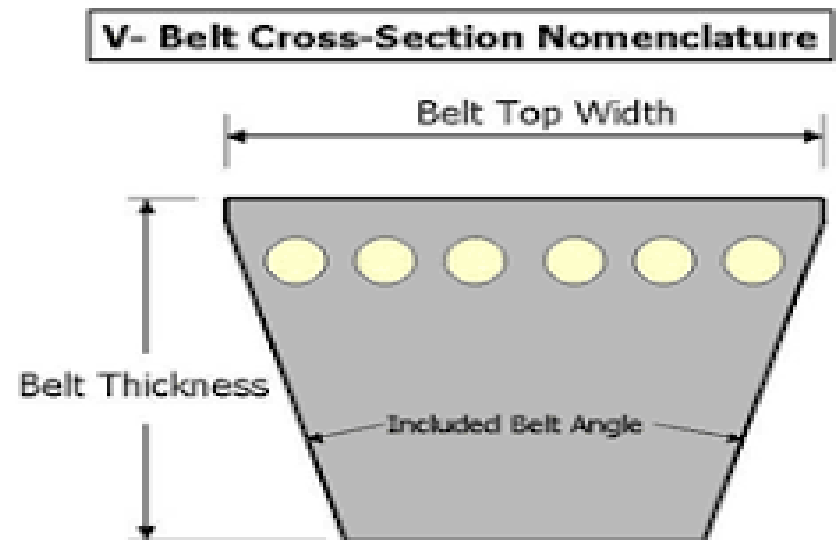
(a) FLAT BELT

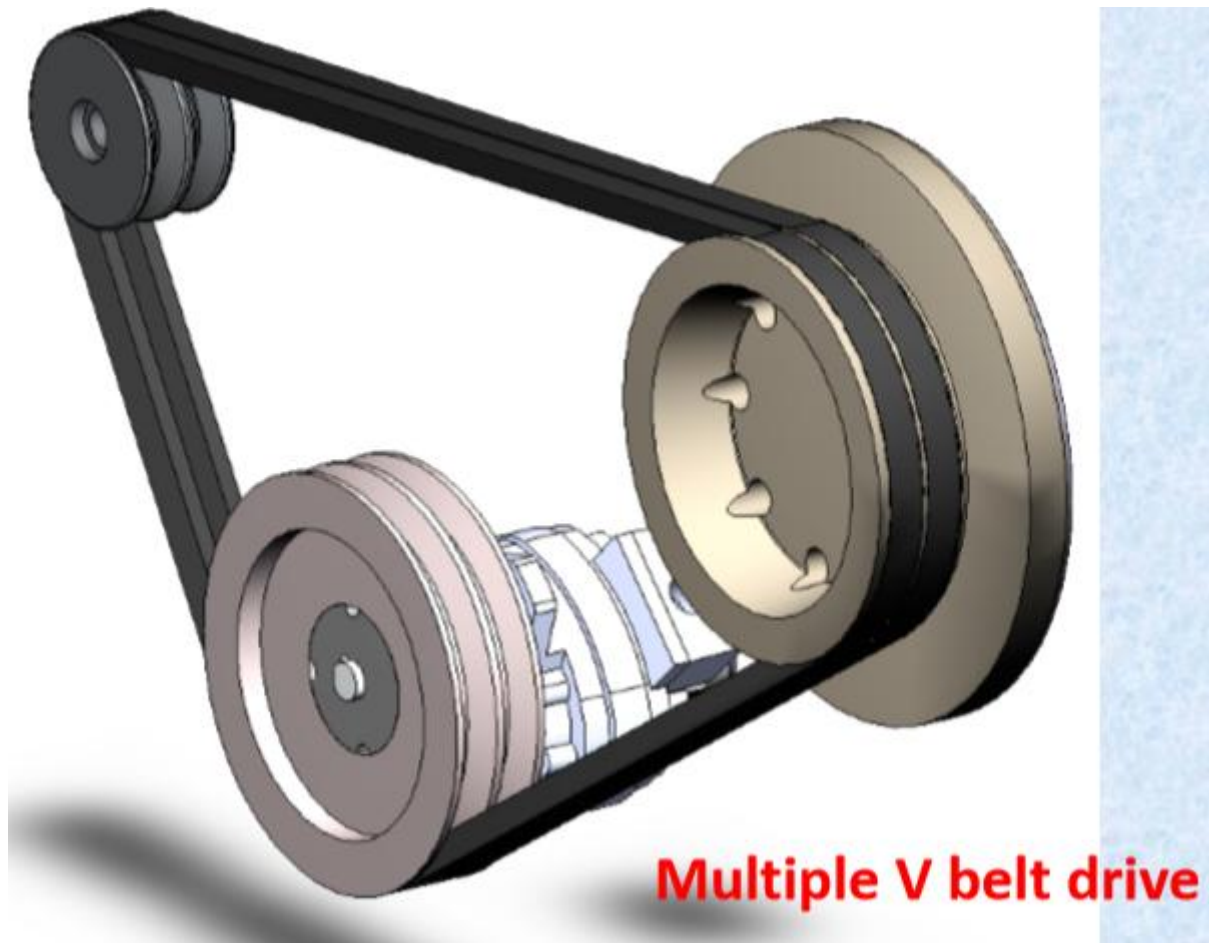
- Flat belt are run on flat pulleys
- They are used in sawmills, conveyors, electrical generators...etc
- Used for moderate amount of power transfer
- Up to 10 m distance
- **Advantages**
- Simple in construction, smooth operation, low maintenance and long life
- Flexible
- **Disadvantages**
- Not positive drives
- Less efficient
- Not suitable for short distances



(b)V BELT

- V belts have trapezoidal cross section
- They run in the V- grooved pulleys
- Multiple V belts are used when the power to be transmitted is large for a single belt
- V belts are used in automobiles and in agricultural purposes





Multiple V belt drive

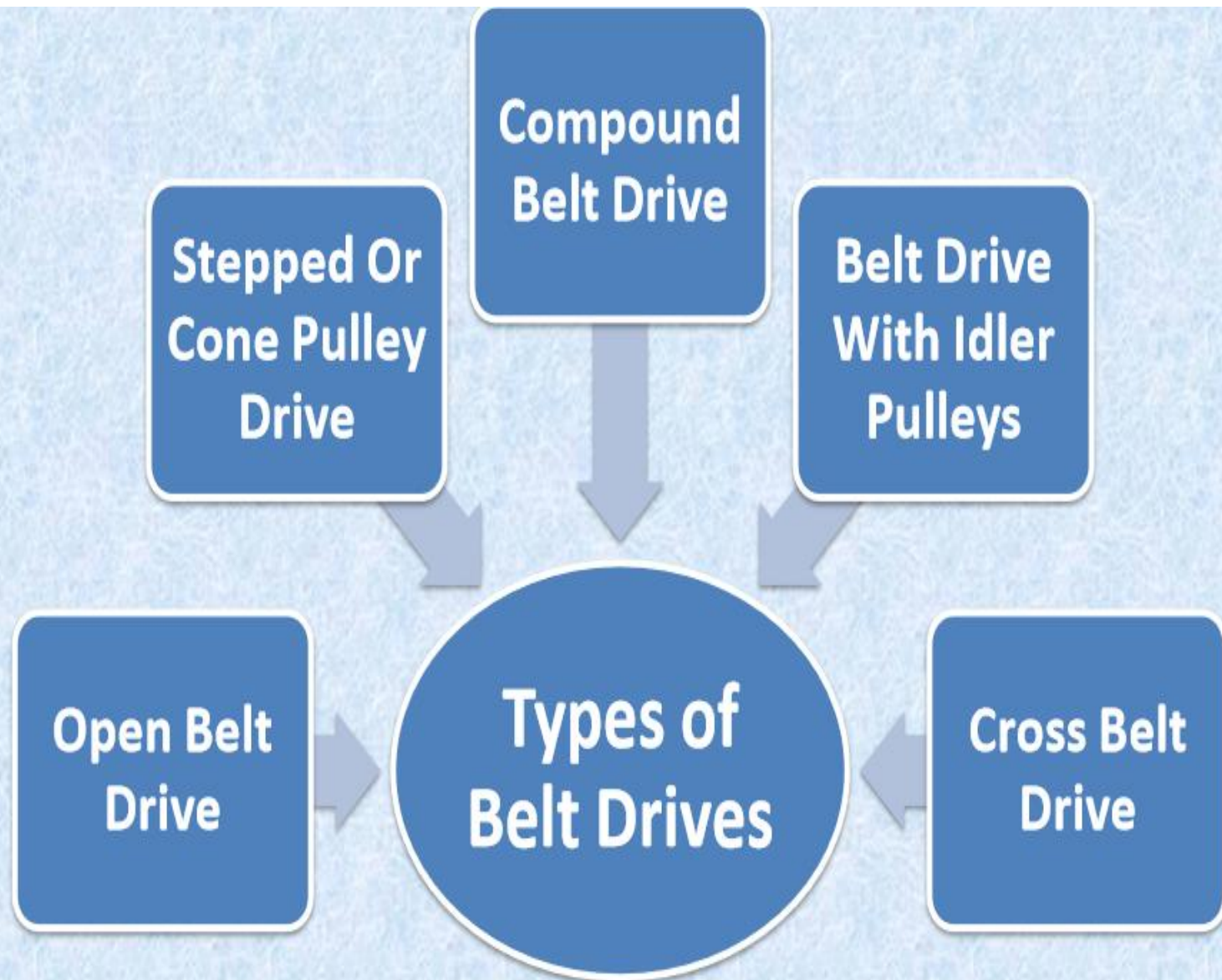
- **Advantages**

- Transmits higher torque than flat belts
 - Suitable for short distance
- Easily installed and removed
 - Slip is negligible
- Operation is quiet
 - Suitable for large speeds

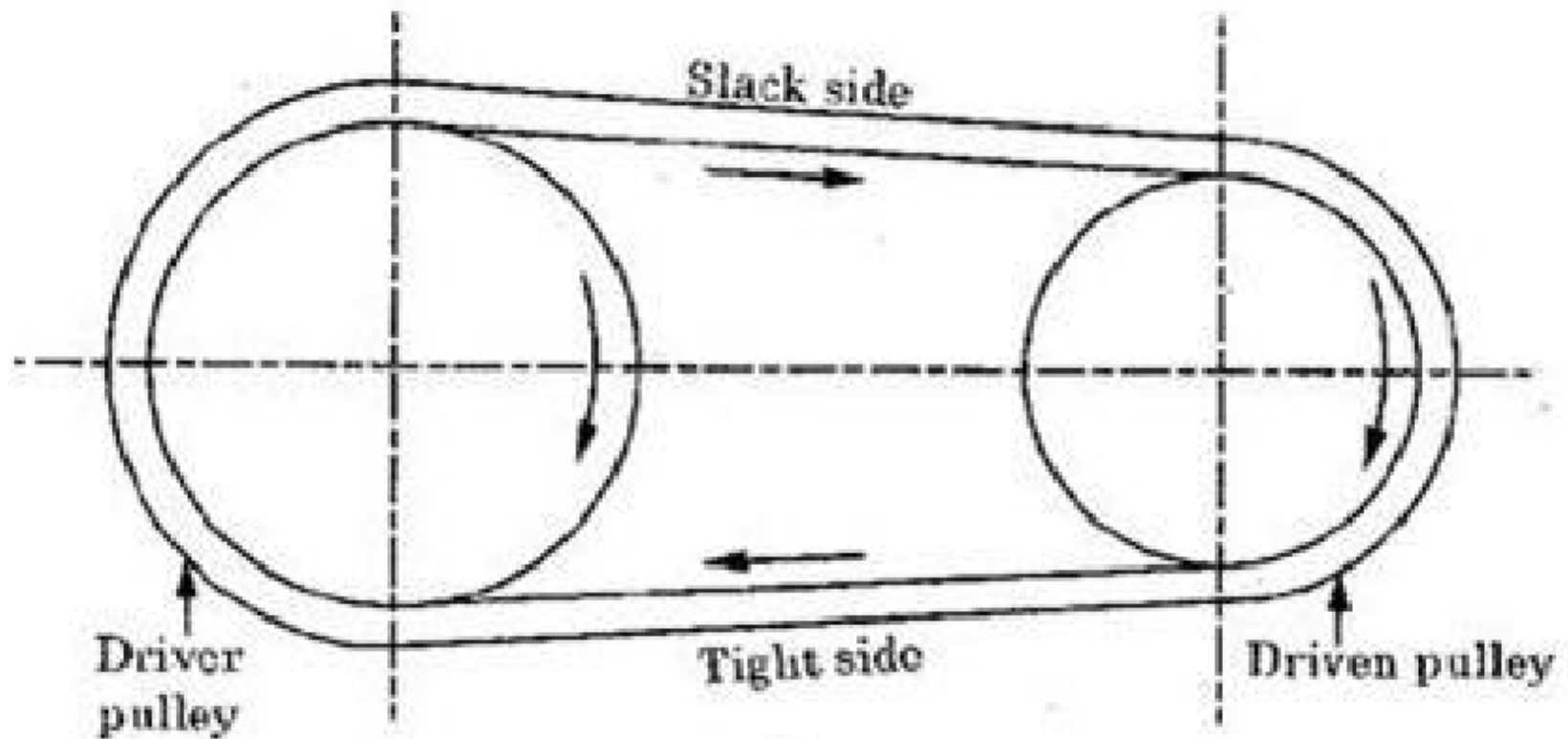
- **Disadvantages**

- Not suitable for large distances
- Costly
 - V belts cannot be repaired
- Construction is complicated

V Belt	Flat Belt
Suitable for shorter distance	Suitable for longer distance
Trapezoidal section	Rectangular section
Frictional grip is more	Frictional grip is less
Power transmitted is more	Power transmitted is less
Velocity ratio is high	Velocity ratio is low
Occurrence of slip is seldom possible	Slip occurs easily



OPEN BELT DRIVE



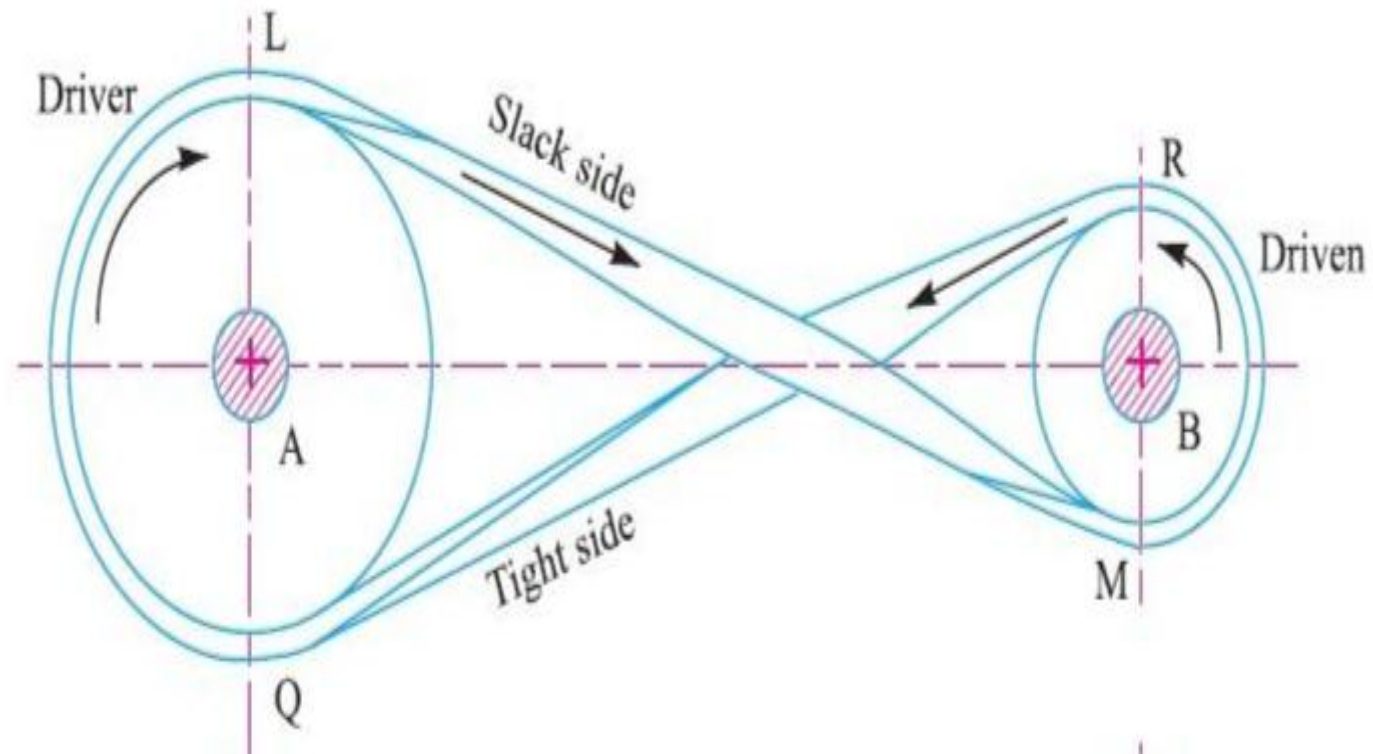
OPEN BELT DRIVE

- It is used to transmit power when the distance between the shafts is large
- Both shafts are in parallel
- Both shafts rotates in the same direction
- When the driver rotates in the clockwise direction, the lower side of the belt is tight and the upper side is slack
- Upper side of the belt is called the **slack** side
- Lower side of the belt is called the **tight** side

CROSS BELT DRIVE

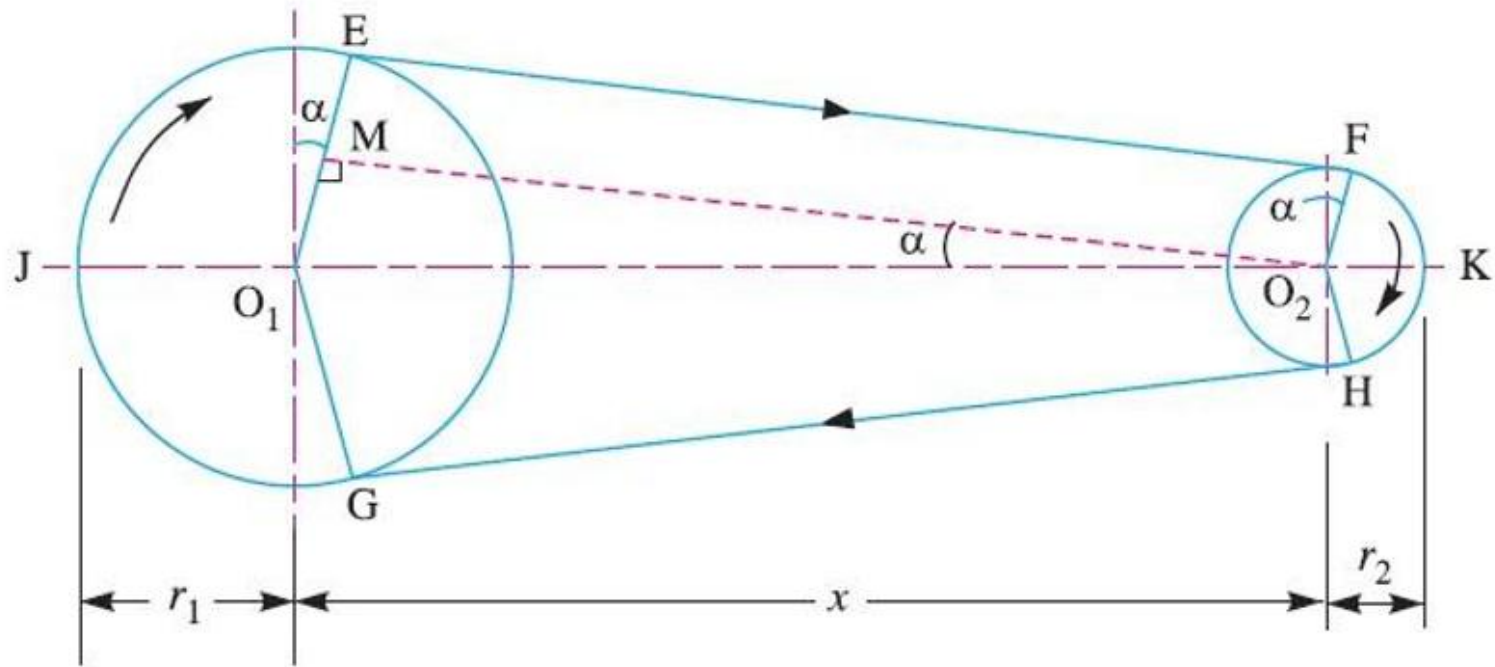
- Shafts are parallel and rotating in the opposite directions
- At the point where the belt crosses, it rubs against itself and wears
- The drive should operate at low velocity

2.Cross Belt Drive



Calculation of length of belt

a) Open Belt Drive



(a) Open belt drive

In open belt drive, the two pulleys rotate in the same direction.

Let x = distance between centres of pulleys

r_1, r_2 = radii of larger and smaller pulleys respectively

L = total length of belt.

The belt leaves the larger pulley at A and C and the smaller pulley at D and F. Through O_2 draw O_2G parallel to CD. Since angle O_1CD is 90° , O_2G will be perpendicular to O_1C . Let angle O_1O_2G be α , then it can be shown that O_1A, O_1C, O_2D and O_2F make angle α with the vertical. Hence the arc of contact between the belt and larger pulley is $(\pi + 2\alpha)$ radian and arc of contact between the belt and smaller pulley is $(\pi - 2\alpha)$ radian. The total length of belt is the sum of the length of belt in contact with larger pulley, length of belt in contact with the smaller pulley and the length of belt not in contact with the pulleys.

$$\text{ie., } L = \text{Arc ABC} + \text{Arc DEF} + (CD + AF)$$

$$\text{Arc ABC} = (\pi + 2\alpha) r_1$$

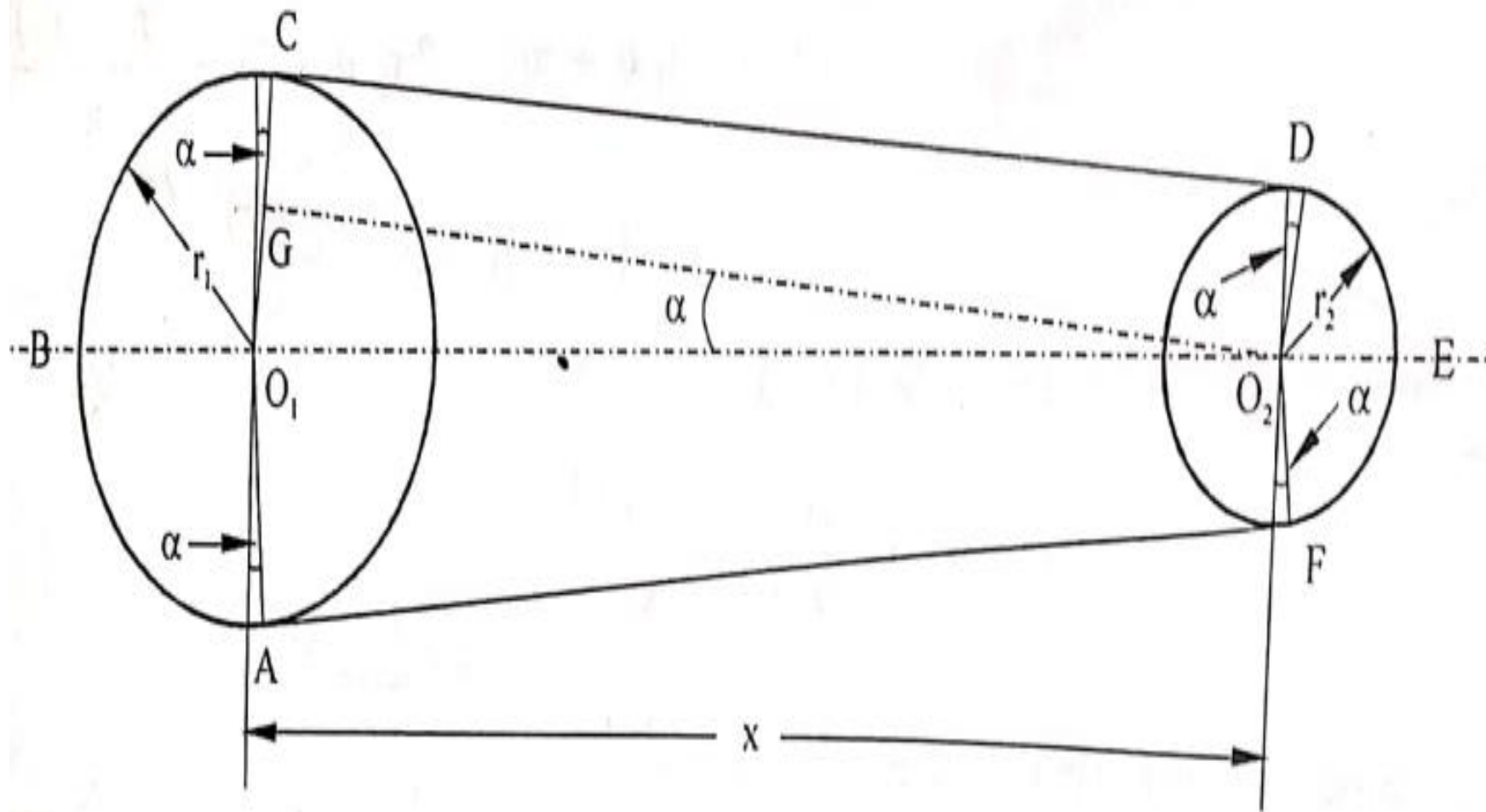


Fig 6.7 Length of Open Belt

$$\text{Arc DEF} = (\pi - 2\alpha) r_2$$

$$(\text{CD} + \text{AF}) = 2 \text{ O}_2\text{G}$$

$$= 2 \sqrt{(\text{O}_1\text{O}_2)^2 - (\text{O}_1\text{G})^2}$$

$$= 2 \sqrt{x^2 - (r_1 - r_2)^2}$$

$$= 2x \sqrt{1 - \left(\frac{r_1 - r_2}{x}\right)^2}$$

Expanding $\sqrt{1 - \left(\frac{r_1 - r_2}{x}\right)^2}$ using Binomial theorem,

$$\sqrt{1 - \left(\frac{r_1 - r_2}{x}\right)^2} = 1 - \frac{1}{2} \left(\frac{r_1 - r_2}{x}\right)^2 + \text{high powers of } \left(\frac{r_1 - r_2}{x}\right)$$

Neglecting higher powers of $\left(\frac{r_1 - r_2}{x}\right)$,

$$\text{CD} + \text{AF} = 2x \left[1 - \frac{1}{2} \left(\frac{r_1 - r_2}{x}\right)^2 \right]$$

$$= 2x - \frac{(r_1 - r_2)^2}{x}$$

$$\therefore L = (\pi + 2\alpha) r_1 + (\pi - 2\alpha) r_2 + 2x - \frac{(r_1 - r_2)^2}{x}$$

$$= \pi r_1 + 2 r_1 \alpha + \pi r_2 - 2 r_2 \alpha + 2x - \frac{(r_1 - r_2)^2}{x}$$

$$= \pi(r_1 + r_2) + 2(r_1 - r_2)\alpha + 2x - \frac{(r_1 - r_2)^2}{x}$$

From the right angled triangle $O_1 O_2 G$,

$$\sin \alpha = \frac{O_1 G}{O_1 O_2} = \frac{(r_1 - r_2)}{x}$$

Since α is very small, $\sin \alpha = \alpha = \frac{(r_1 - r_2)}{x}$

Substituting this value of α in the expression for L ,

$$L = \pi(r_1 + r_2) + 2(r_1 - r_2) \frac{(r_1 - r_2)}{x} + 2x - \frac{(r_1 - r_2)^2}{x}$$

$$= \pi(r_1 + r_2) + 2 \frac{(r_1 - r_2)^2}{x} + 2x - \frac{(r_1 - r_2)^2}{x}$$

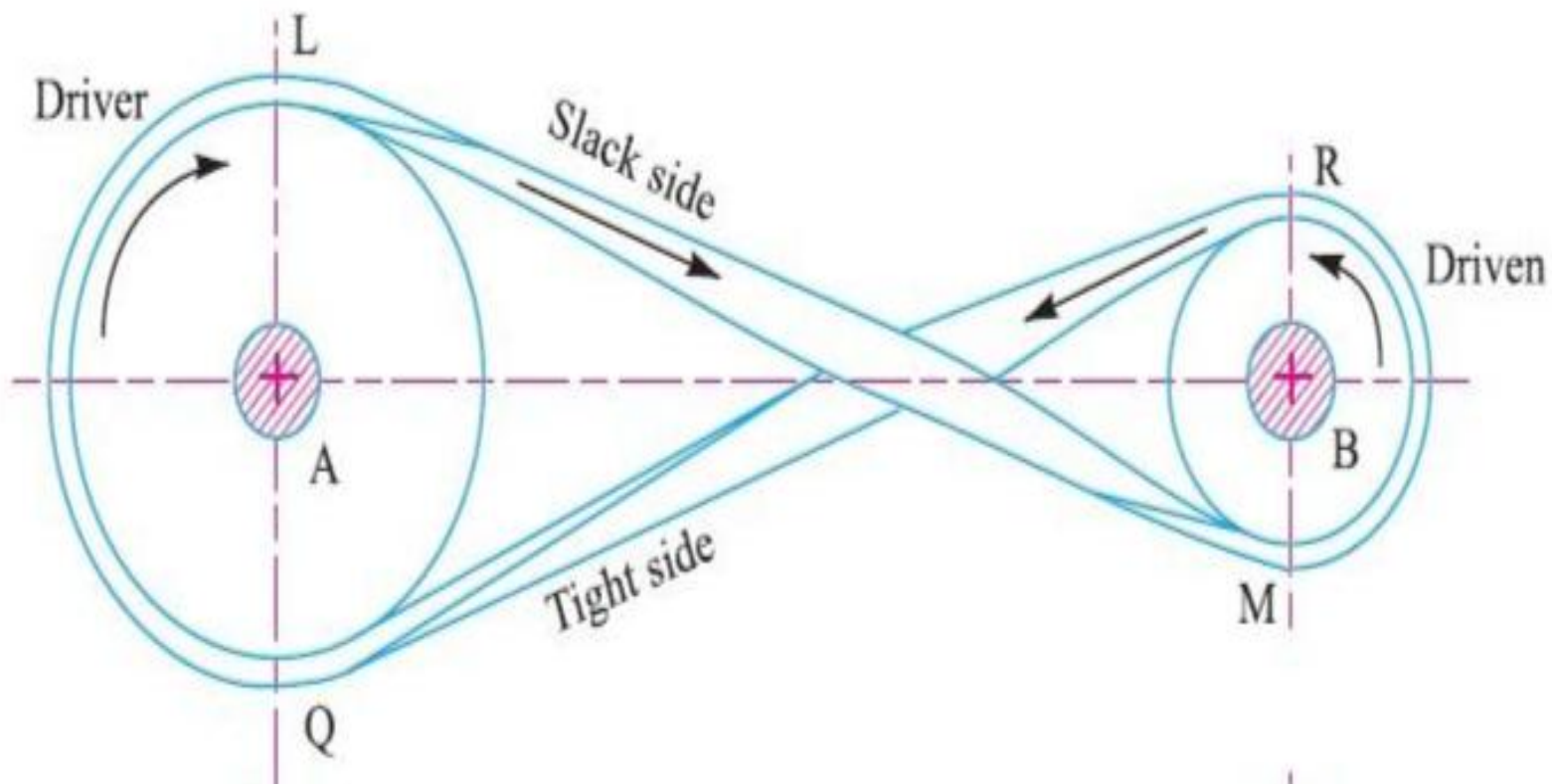
$$L = \pi(r_1 + r_2) + \frac{(r_1 - r_2)^2}{x} + 2x \dots \dots \dots (i)$$

- X = distance between centers of pulleys
- r_1, r_2 = radii of larger and smaller pulleys respectively
- L = total length of belt

$$L = \pi(r_1 + r_2) + \frac{(r_1 - r_2)^2}{X} + 2X \dots\dots$$

$$\alpha = (r_1 - r_2)/X$$

2. Cross Belt Drive



- X = distance between centers of pulleys
- r_1, r_2 = radii of larger and smaller pulleys respectively
- L = total length of belt

$$L = \pi(r_1 + r_2) + \frac{(r_1 + r_2)^2}{X} + 2X \dots\dots$$

$$\alpha = (r_1 + r_2)/X$$

Ratio of Belt Tensions

- T_1 = tight side tension
- T_2 = slack side tension
- Θ = angle of contact between the belt and pulley in radians
- μ = coefficient of friction between the belt and pulley

$$T_1/T_2 = e^{\mu \Theta}$$

Power Transmitted By Belt Drive

- T_1 = tight side tension, N
- T_2 = slack side tension, N
- r = radius of pulley, m
- d = diameter of pulley, m
- ω = angular velocity, rad/sec
- V = velocity of belt, m/s
- N = speed of pulley(rotations per minute, rpm)
- If **d** is the diameter of **driver pulley** then **N** should be the speed of **driver pulley** and if **d** is the diameter of **driven pulley** then **N** should be the speed of **driven pulley**

- Torque exerted on the pulley = $(T_1 - T_2)r$
- Work done per sec = $(T_1 - T_2)r \omega$

- **Power transmitted = $(T_1 - T_2)V$**

- $V = r \omega$
- $\omega = 2\pi N/60$

- **Power transmitted = $[(T_1 - T_2) \pi dN]/60$, Watts**

Velocity Ratio of a Belt Drive

Definition of Velocity Ratio for Belt Drives. (Speed Ratio)

The velocity ratio of a belt drive is defined as the ratio of the speed of the driven pulley to the speed of the driving pulley.

- Velocity ratio, $i = \frac{\text{speed of driven shaft}}{\text{speed of driver shaft}}$

Also, Velocity ratio, $i = \frac{\text{diameter of driver pulley}}{\text{diameter of driven pulley}}$

$$i = N_2/N_1 = d_1/d_2$$

- N_2 = speed of driven shaft
- N_1 = speed of driver shaft
- d_1 = diameter of driver pulley
- d_2 = diameter of driven pulley

- If the thickness of the belt, t is also considered, then

$$i = N_2/N_1 = (d_1+t)/(d_2+t)$$

Example 15.4. A crossed belt is used for connecting two parallel shafts 5 metres apart. The pulley diameters of the driver and driven shafts are 300 mm and 500 mm respectively. Find out the length of belt required and the angle of contact of the belt with the larger pulley. How much length of belt has to be reduced, in order to change over to an open belt instead of crossed belt.

Solution. Given : $d_1 = 300$ mm, $d_2 = 500$ mm, $x = 5000$ mm,

We have,

$$r_1 = 150 \text{ mm}$$

$$r_2 = 250 \text{ mm}$$

i) Angle of contact with the larger pulley $= \pi + 2\alpha$

$$\alpha = \sin \alpha = \frac{r_2 + r_1}{x} = \frac{250 + 150}{5000} = 0.08 \text{ radians}$$

$$\therefore \text{Angle of contact} = \pi + 2\alpha = 3.3016 \text{ radians} = 189.17^\circ$$

$$\begin{aligned} \text{Length of crossed belt} &= \pi(r_2 + r_1) + 2x + \frac{(r_2 + r_1)^2}{x} \\ &= 11288.64 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{ii) Length of open belt} &= \pi(r_2 + r_1) + 2x + \frac{(r_2 - r_1)^2}{x} \\ &= 11258.64 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Length to be reduced} &= 11288.64 - 11258.64 \\ &= 30 \text{ mm} \end{aligned}$$

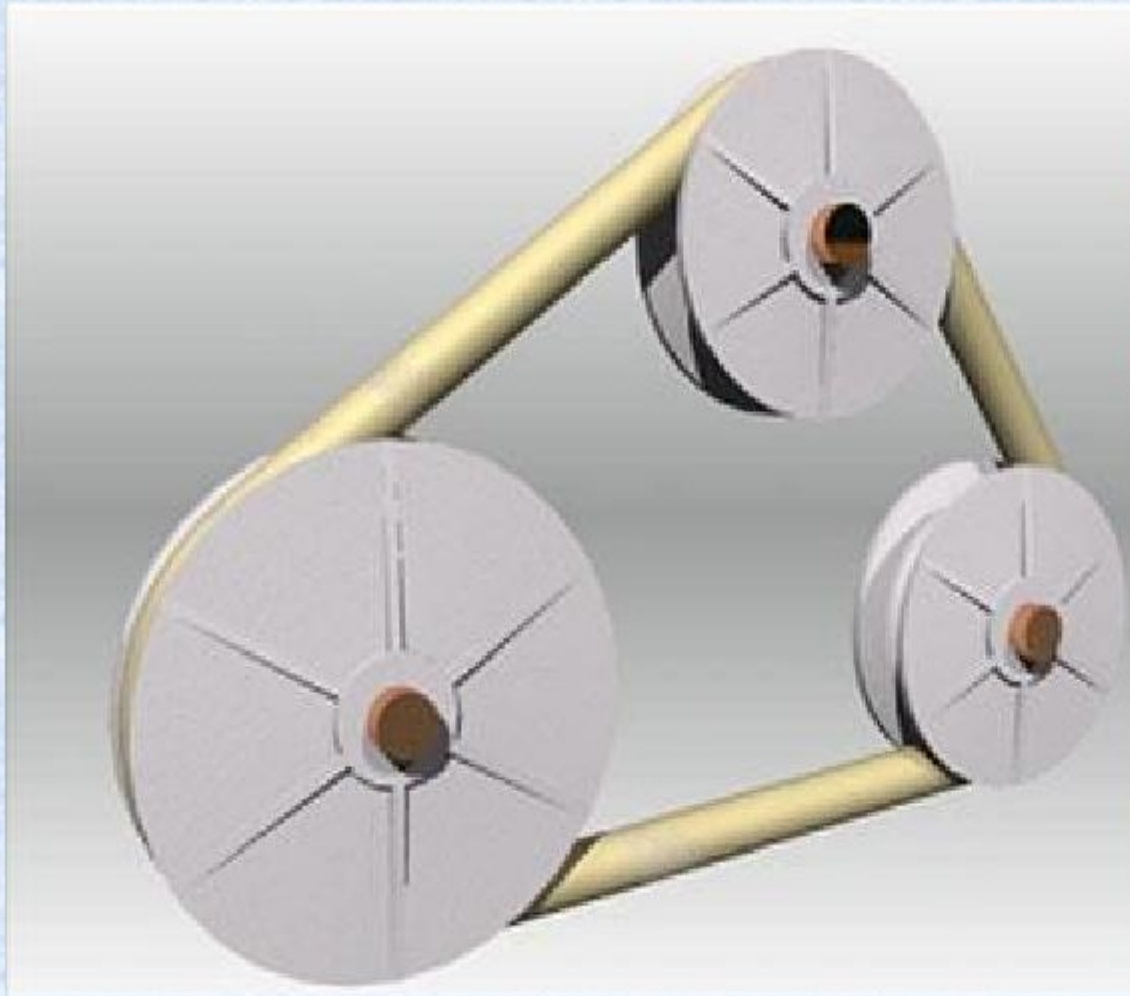
Slip of Belt Drive

- Slip is measured as a difference between the linear speeds of the pulley rim and the belt over it
- Generally it is expressed as a percentage

Slip occurs due to

- Friction between the belt and pulley decreases
- When the smoothness of pulley surface is high
- Difference between tensions in the tight and slack sides are very large

ROPE DRIVE



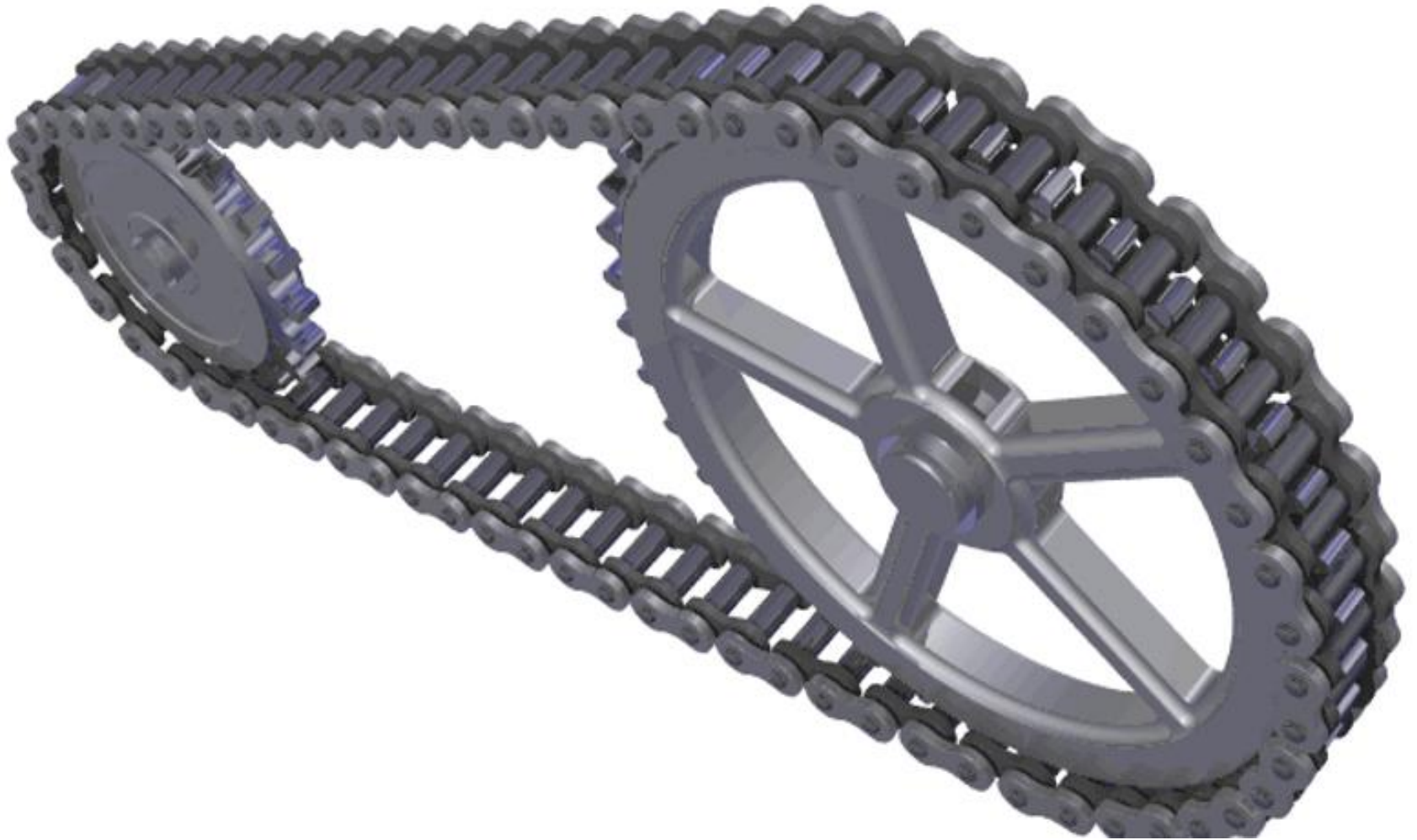
ROPE DRIVE

- Rope drives are used for transmission of large amount of power over long distances
- For a rope drive, groove angle of the pulley is usually 45°
- Fibre ropes and wire ropes are used

Advantages

- Smooth and silent operation
 - Less weight
 - Longer life
 - Efficiency is high
- More reliable
 - Low cost

CHAIN DRIVE



CHAIN DRIVE

- Chain drive consists of an endless chain running over special profile toothed wheels called sprockets
- One of the sprockets will be the driver and the other driven
- Smaller sprocket is called pinion and the bigger one is called wheel

Advantages

- It is a non-slip drive
- Very high efficiency
- Occupies less space

Disadvantages

- High cost
- More weight
- Needs accurate mountings
- Lubrication is critical



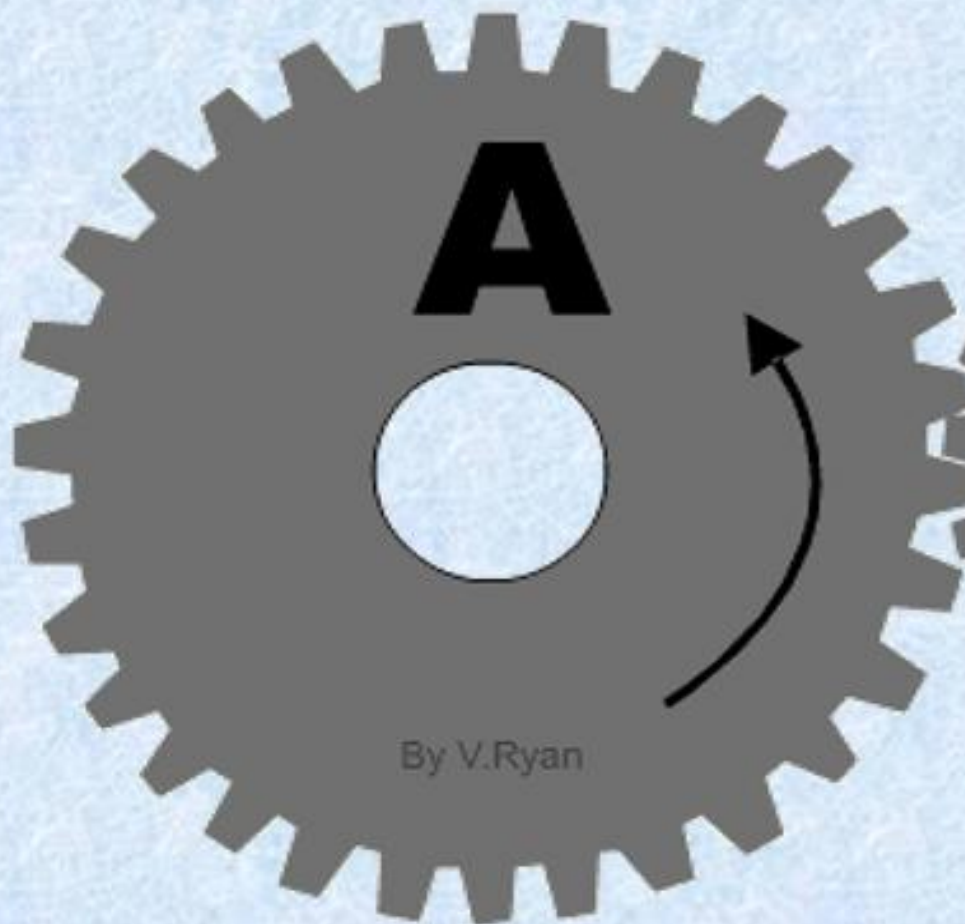
CHAIN

SPROCKET

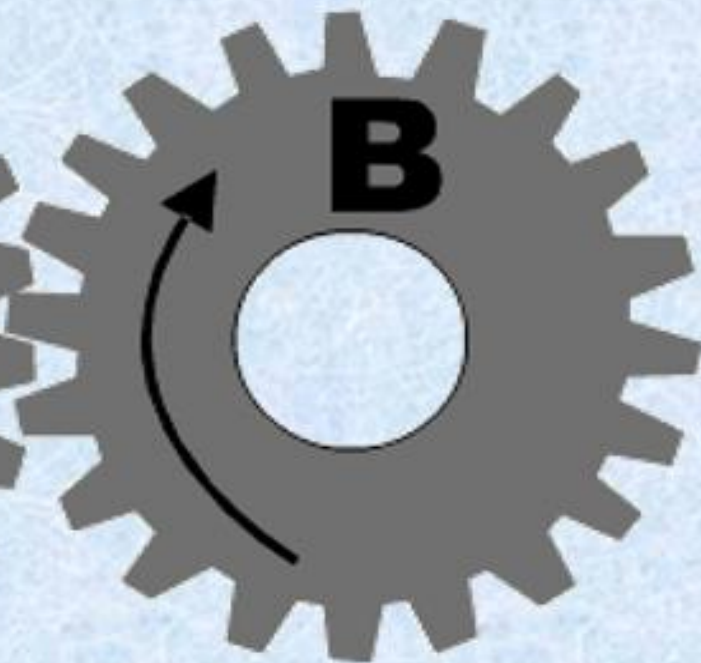


GEAR DRIVE



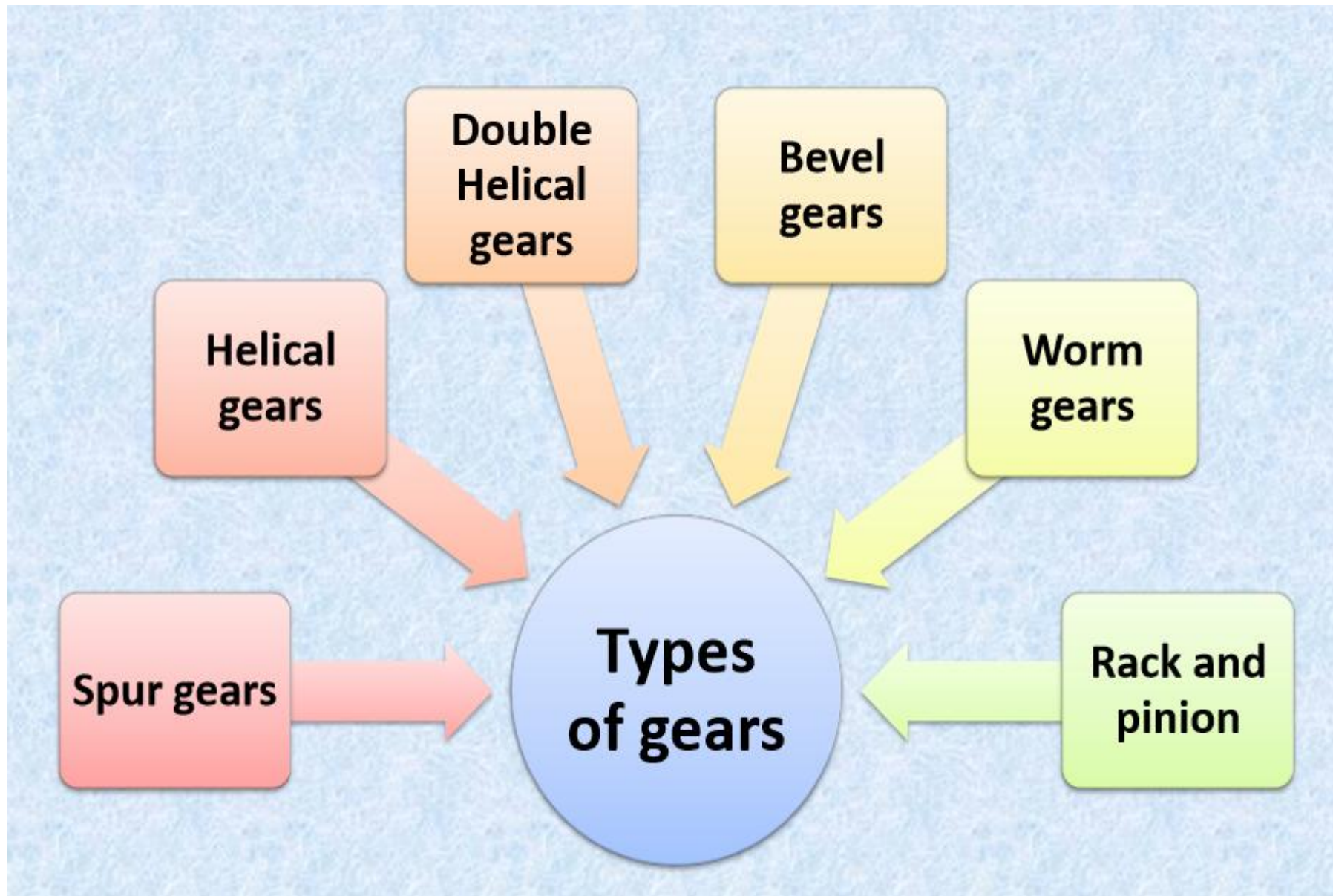


DRIVER



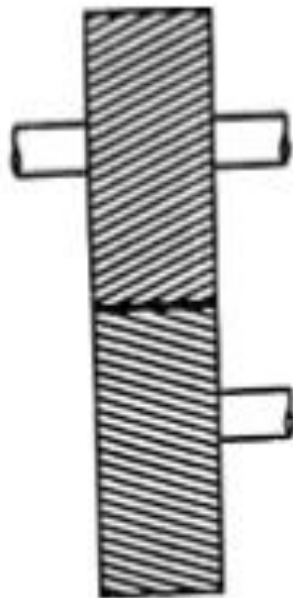
DRIVEN

- A gear is a toothed wheel
- One gear is mounted on the driving shaft and another one on the driven shaft, their teeth meshing with each other
- It is a positive drive(no slip)
- The axes of the shafts may be parallel or non-parallel
- When two gears of different sizes mesh, the smaller one is called pinion and the larger one is called gear
- When pinion(smaller gear) is the driver, output speed(driver speed) decreases and torque increases
- When the gear(larger gear) is the driver, output speed(driver speed) increases and torque decreases

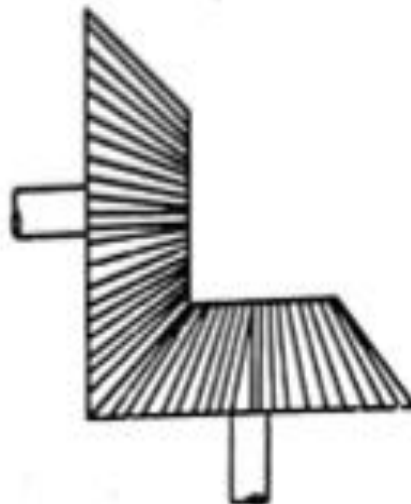




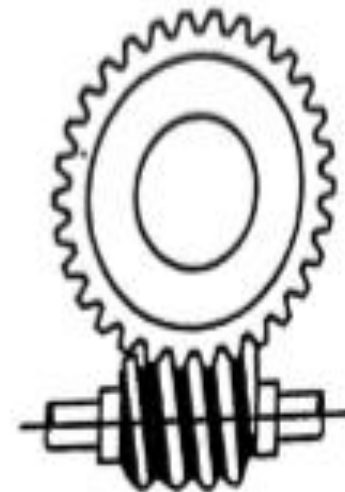
(a) spur gear



(b) Helical gear



(c) Bevel gear



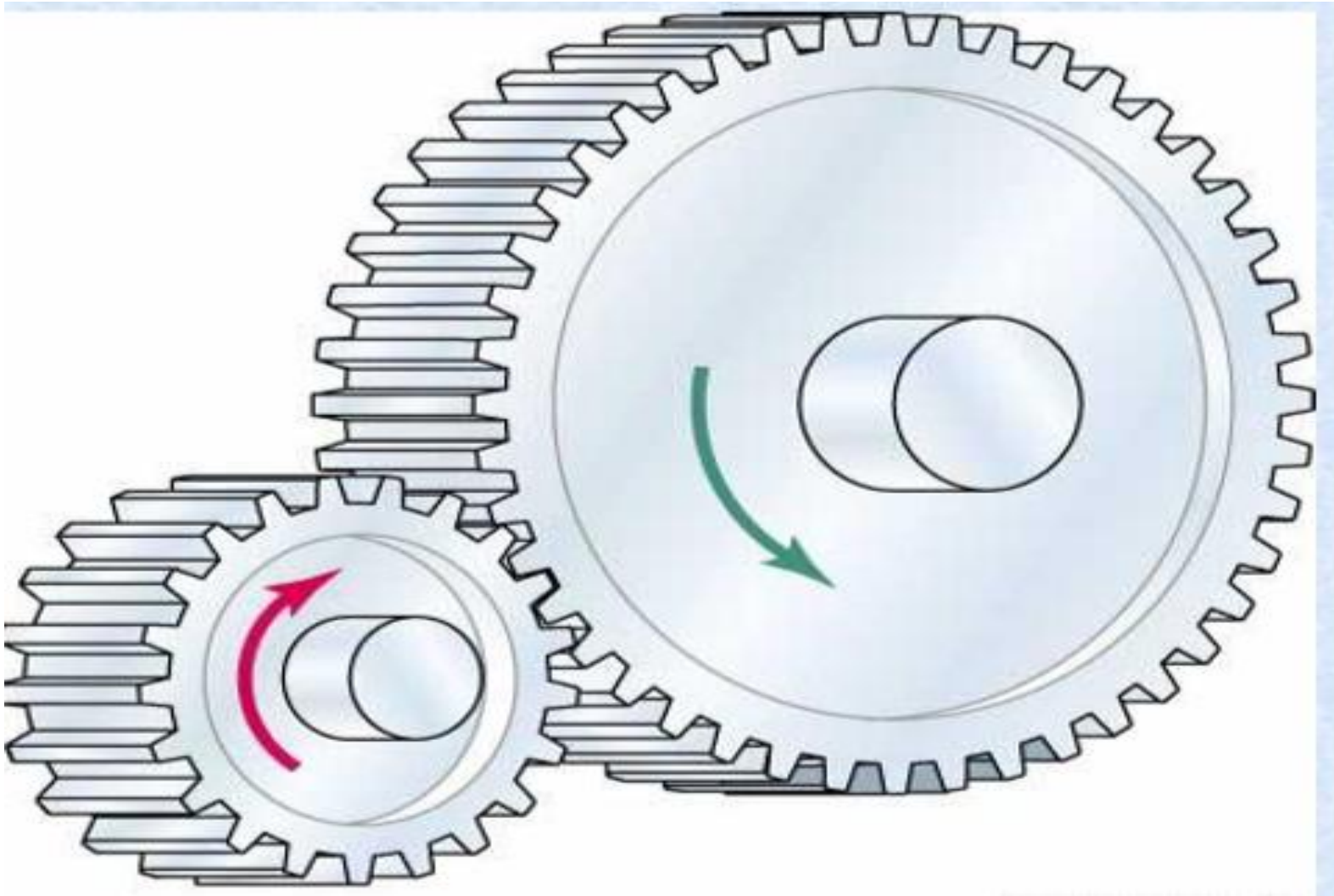
(d) Worm gear

Fig.15.12 Types of gears

1.SPUR GEAR

- SPUR gears are those which have teeth cut parallel to the axis of the shaft
- Spur gears are used to transmit power between parallel shafts
- They are used in high speed and high load applications
- Two types External gear and internal gear
- Internal gears have teeth on the inner periphery
- Two gears rotate in same direction
- Pinion or smaller gear is inside the spur gear
- Internal gears are used in heavy duty tractors, where much torque is required
- External gears have teeth on the outer surfaces and the two shafts rotate in opposite directions

SPUR GEAR



2.HELICAL GEAR



2.HELICAL GEAR

- In helical gear, the teeth cut on the periphery are of helical screw form
- Helical tooth is inclined at an angle to the axis of the shaft
- Helical gears are used to transmit power between parallel shafts
- The two shafts rotate in opposite directions
- They have higher load carrying capacity
- They operate smoother and quieter than spur gears

3.Herringbone Gear or Double-Helical Gears



- **Herringbone gears** have opposing helical teeth which nullify two axial thrusts
- Load carrying capacity is very high
- These gears are used to transmit power between two parallel shafts at high speeds
- The two shafts rotate in opposite directions

4.BEVEL GEAR



BEVEL GEAR

- Bevel gears are used to connect two non-parallel shafts with intersecting axes
- Teeth of these gears are formed on a conical surface
- These gears are used to transmit power between two shafts at any angle, generally the shafts are at right angles
- They are used to slow speed applications

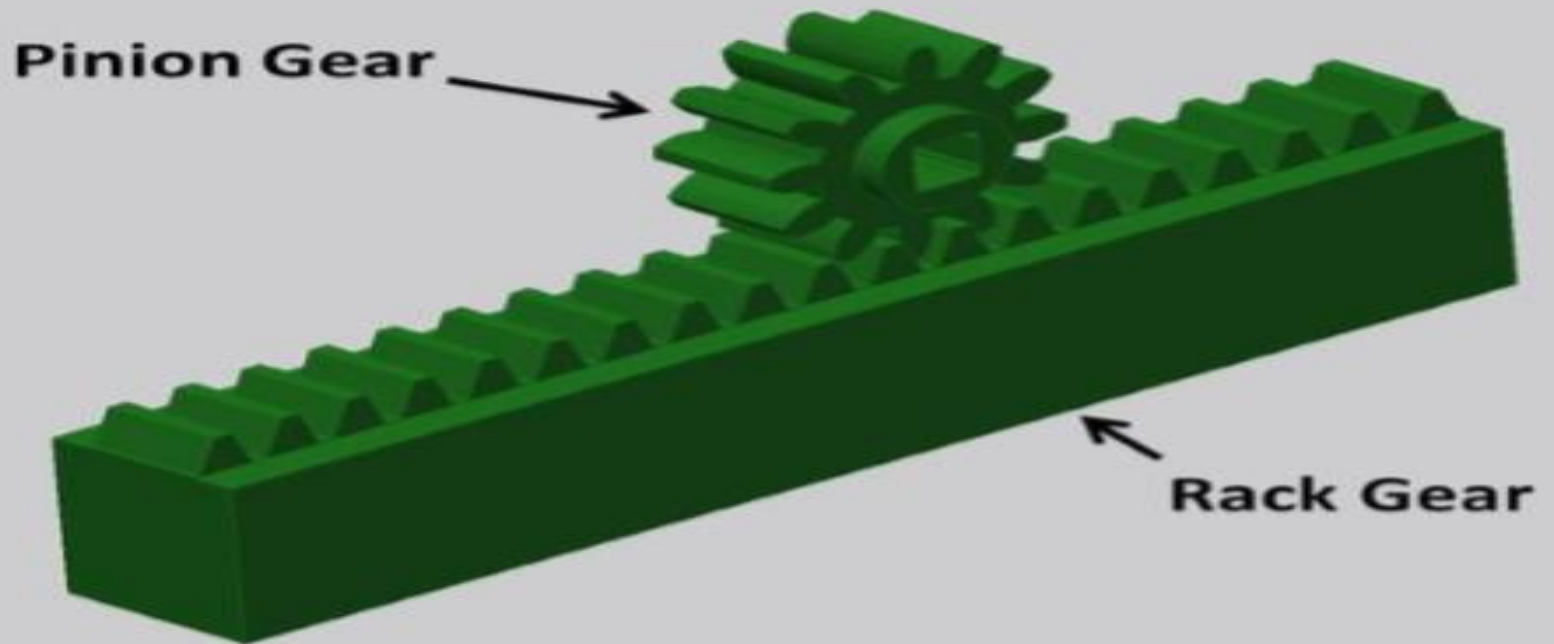
5.WORM GEAR



WORM GEAR

- Worm gears are used for power transmission between non-intersecting shafts that are generally at right angles to each other
- Worm gearing consists of worm and worm wheel
- Worm is a threaded screw and is used as the driver
- Worm wheel is a toothed wheel
- Teeth of the worm wheel remain engaged with the threads of the worm

6.RACK AND PINION



Rack and pinion gears are used to convert rotation(pinion) into linear motion(rack) or vice versa

Advantages of gear drive

- Smooth and reliable
- Transmits more power

Disadvantages

- Not suitable for large distances
 - Needs lubrication
- Maintenance cost is high
 - Power loss due to friction
- Production cost is high